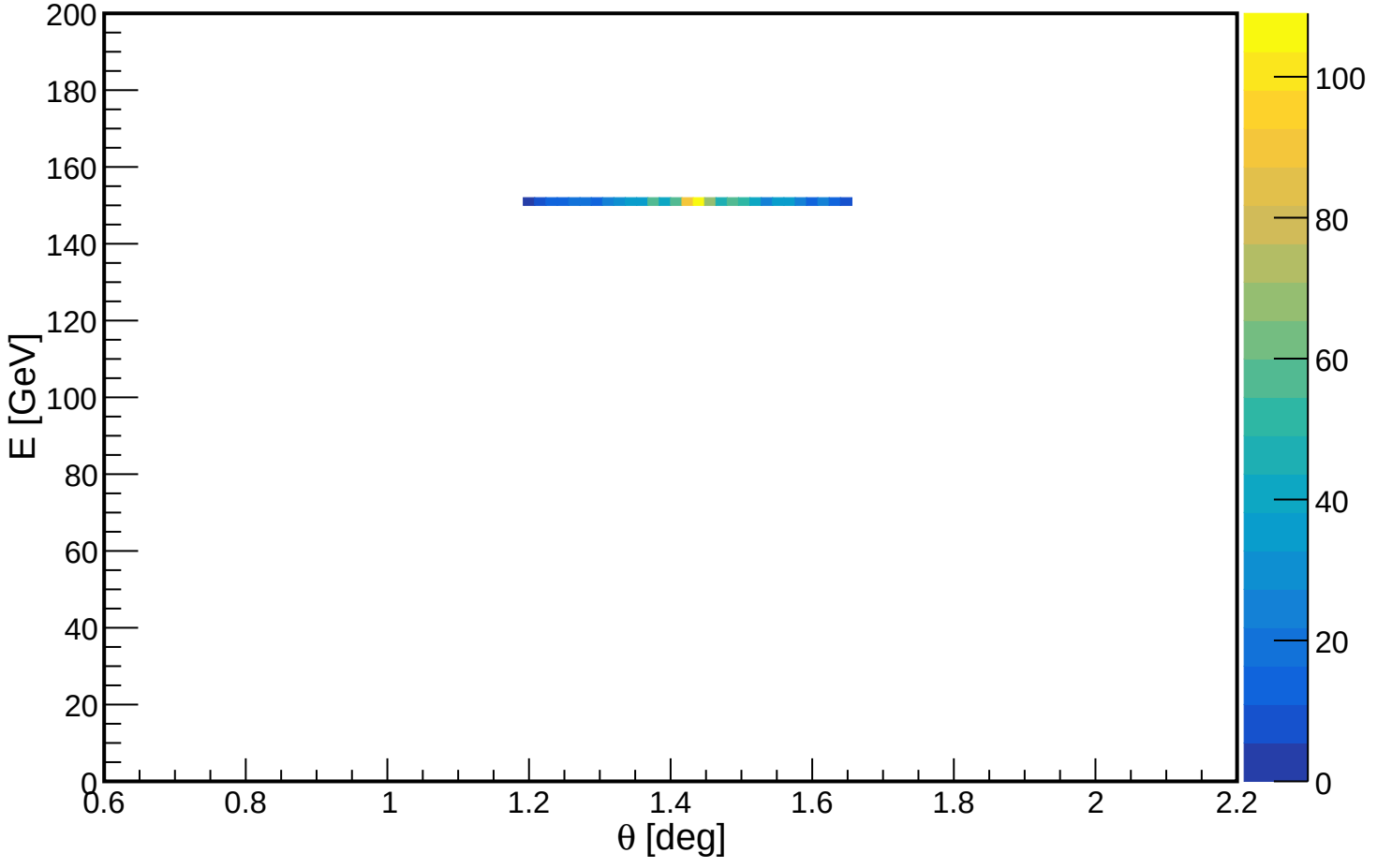
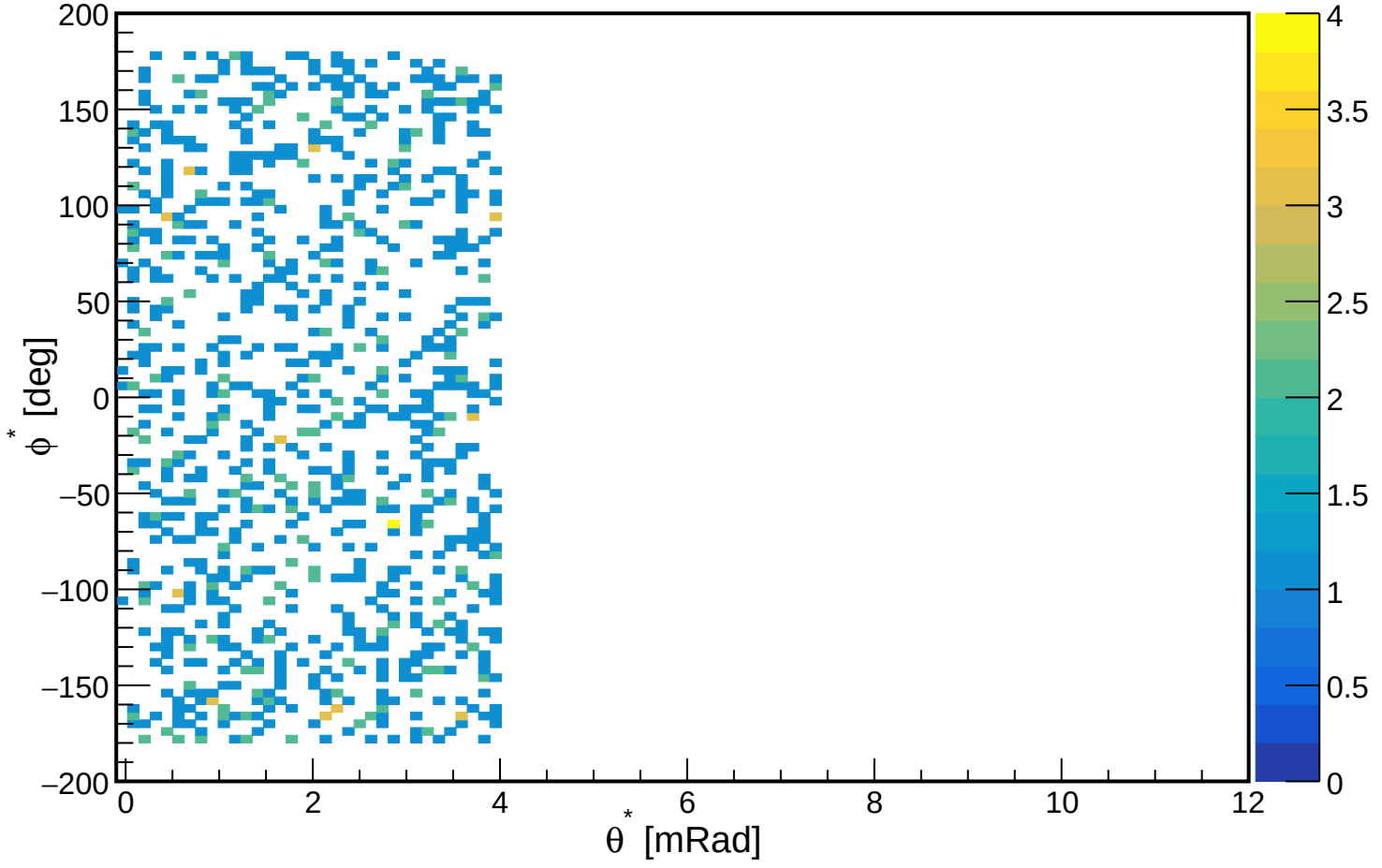
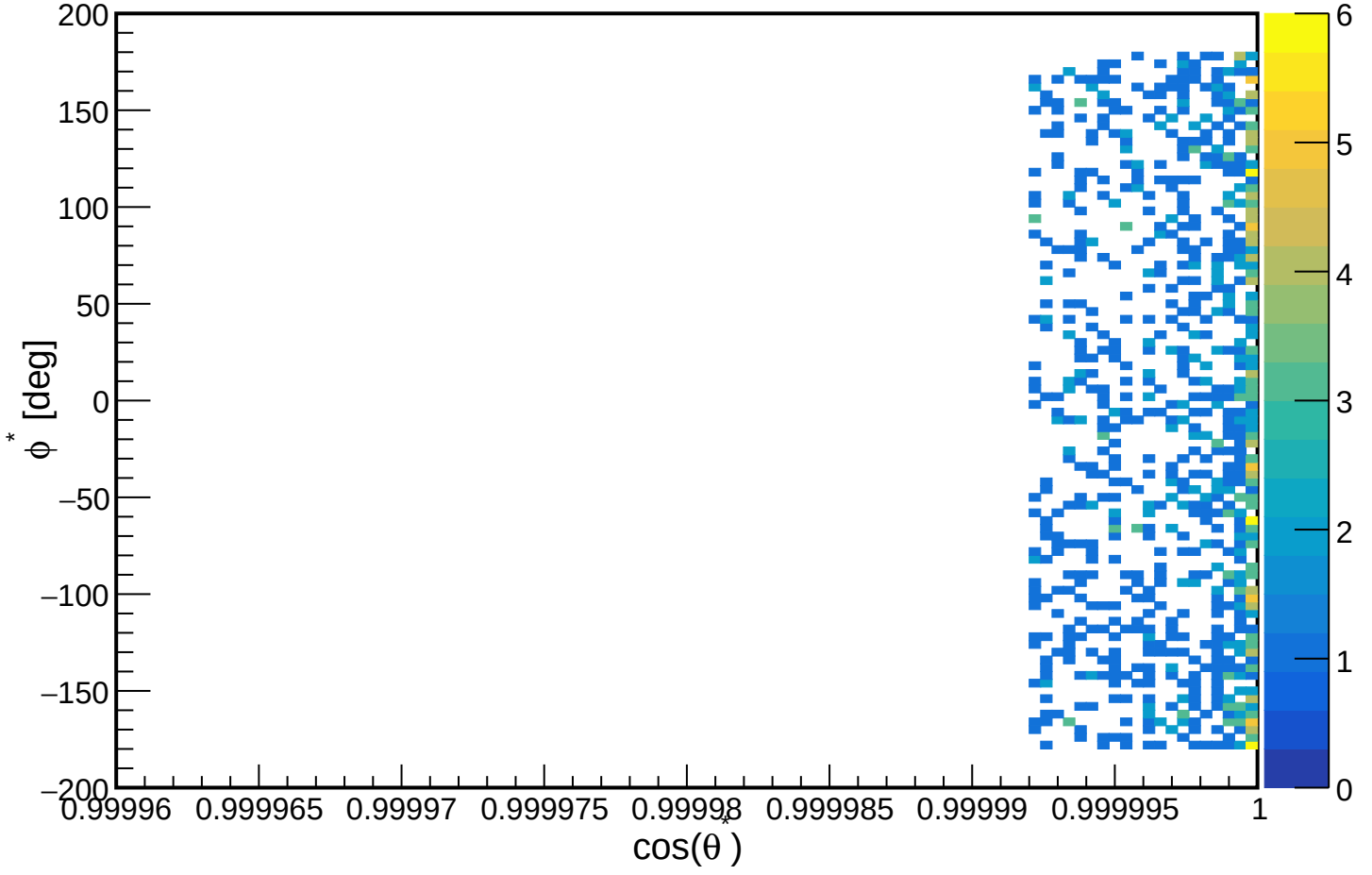
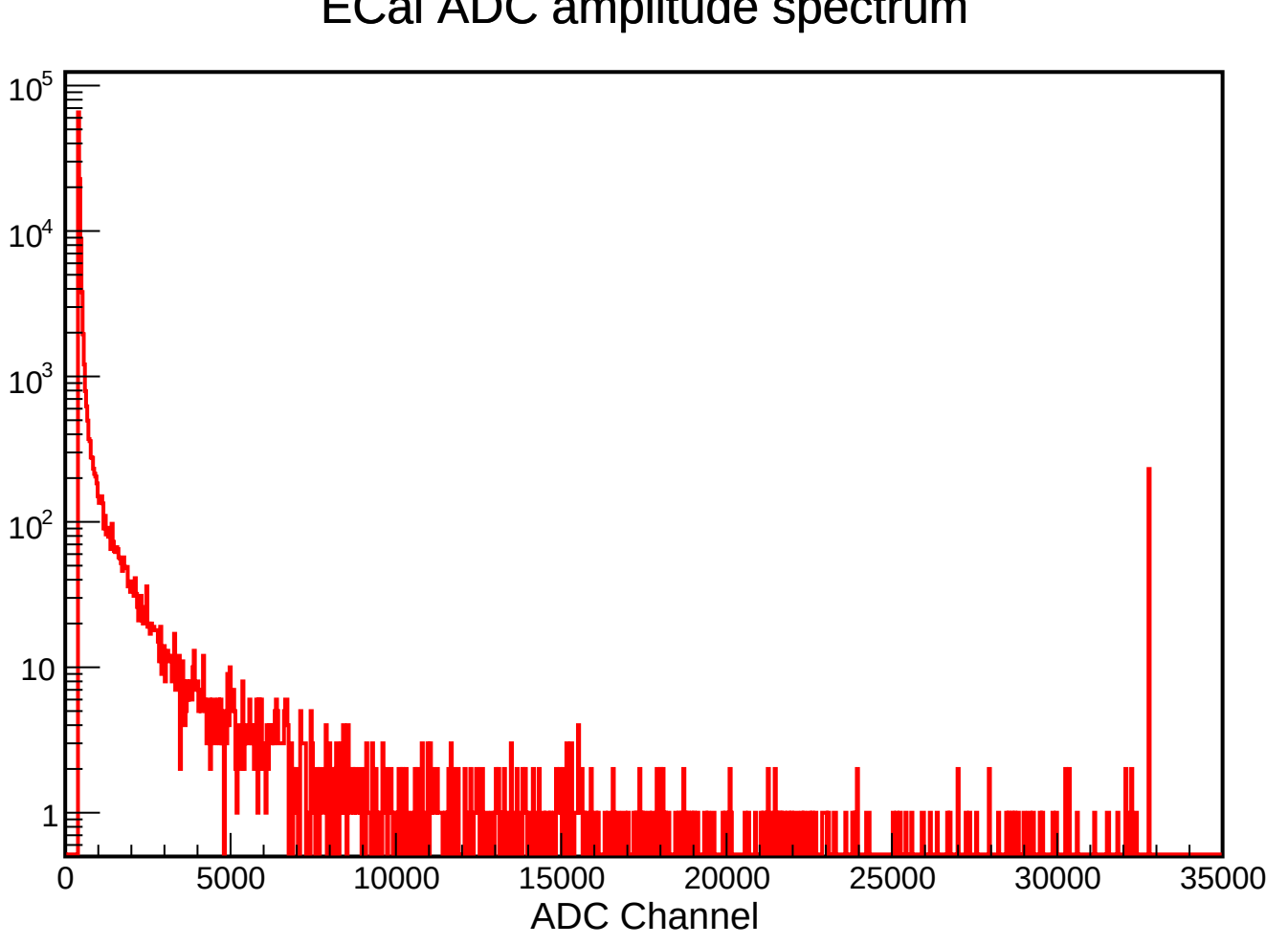
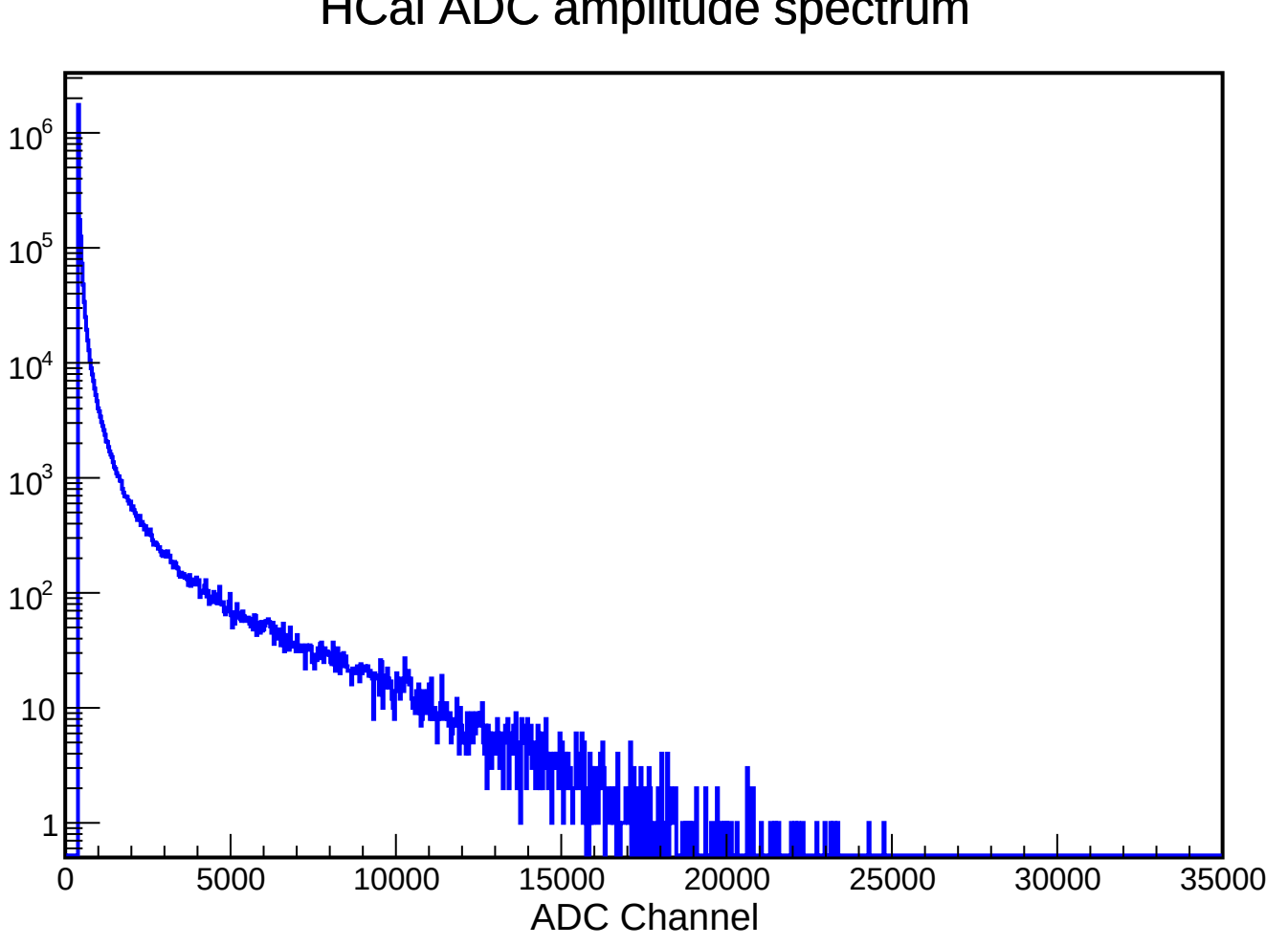

Neutron true energy vs. polar angle
E [GeV]
θ [deg]
0
20
40
60
80
100
120
140
160
180
200
0.6
0.8
1
1.2
1.4
1.6
1.8
2
2.2
0
20
40
60
80
100

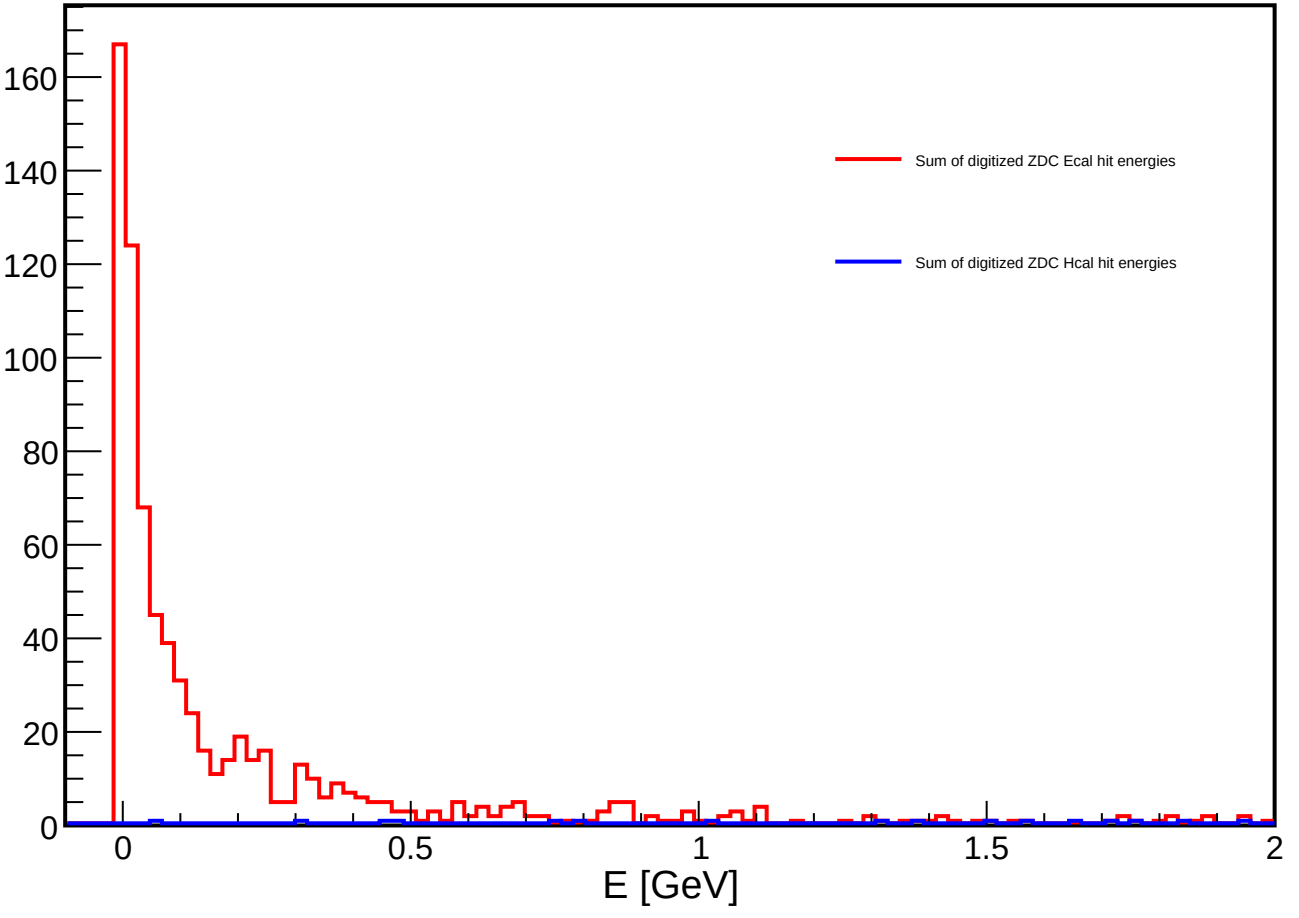

Neutron true azimuthal angle vs. polar angle around p axis
200
150
100
50
0
−50
−100
−150
−200
$\phi^*$ [deg]
0
2
4
6
8
10
12
$\theta^*$ [mRad]
4
3.5
3
2.5
2
1.5
1
0.5
0


Neutron true azimuthal angle vs. cosine of polar angle around p axis
200
150
100
50
0
−50
−100
−150
−200
$\phi^*$ [deg]
0.99996
0.999965
0.99997
0.999975
0.99998
0.999985
0.99999
0.999995
1
$\cos(\theta^*)$
6
5
4
3
2
1
0

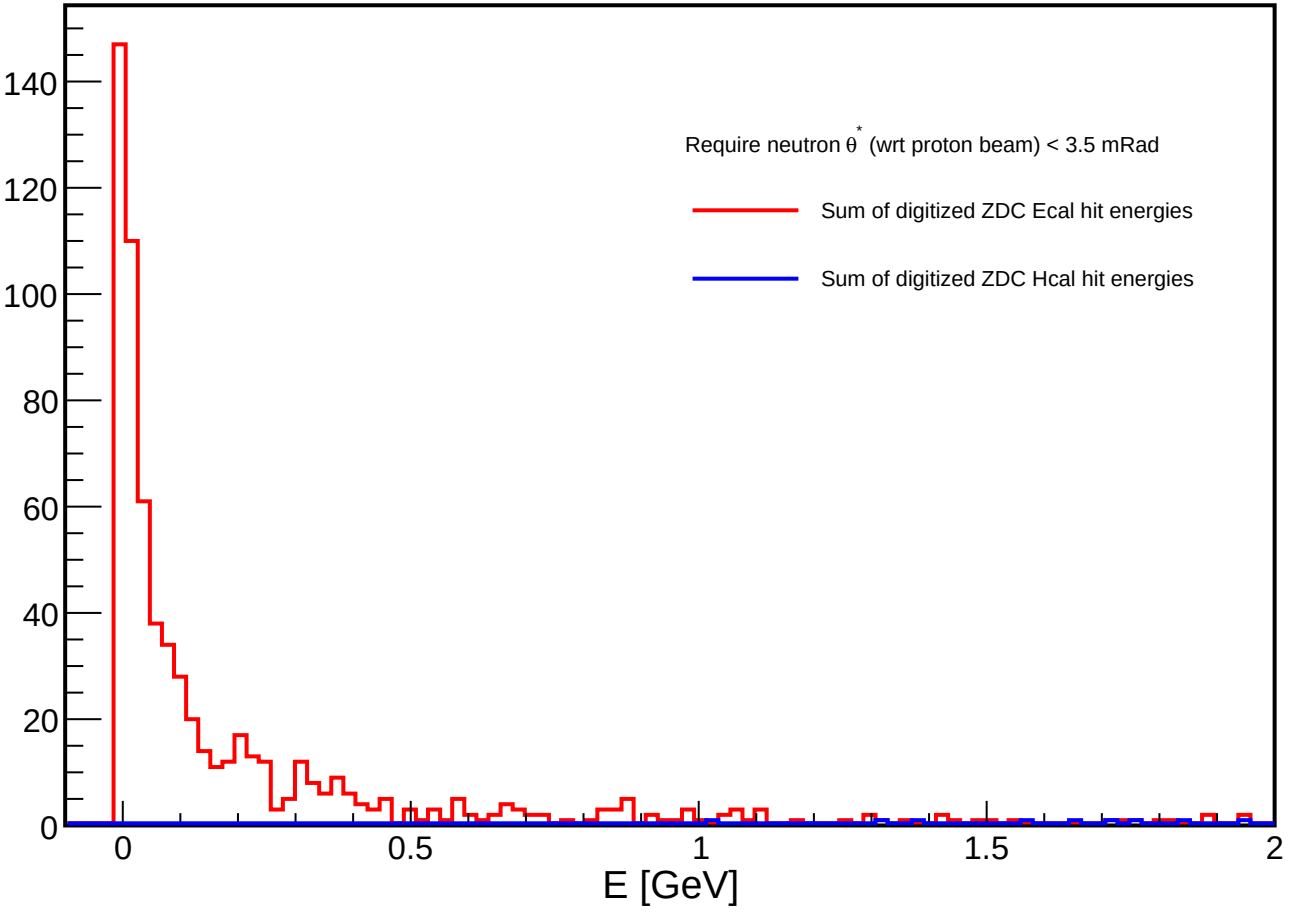

ECal ADC amplitude spectrum
$10^5$
$10^4$
$10^3$
$10^2$
10
1
0
5000
10000
15000
20000
25000
30000
35000
ADC Channel

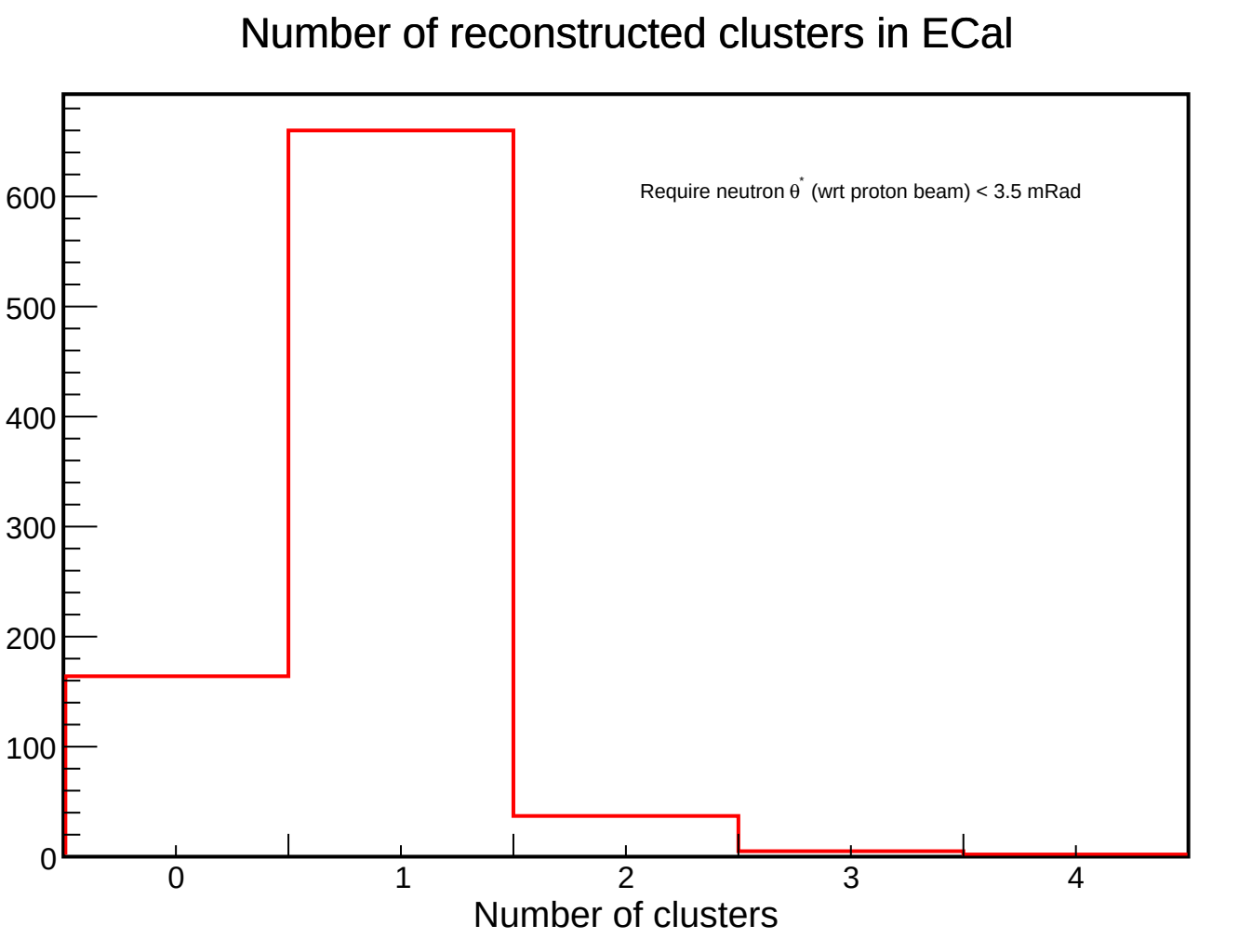

HCal ADC amplitude spectrum
$10^6$
$10^5$
$10^4$
$10^3$
$10^2$
10
1
0
5000
10000
15000
20000
25000
30000
35000
ADC Channel

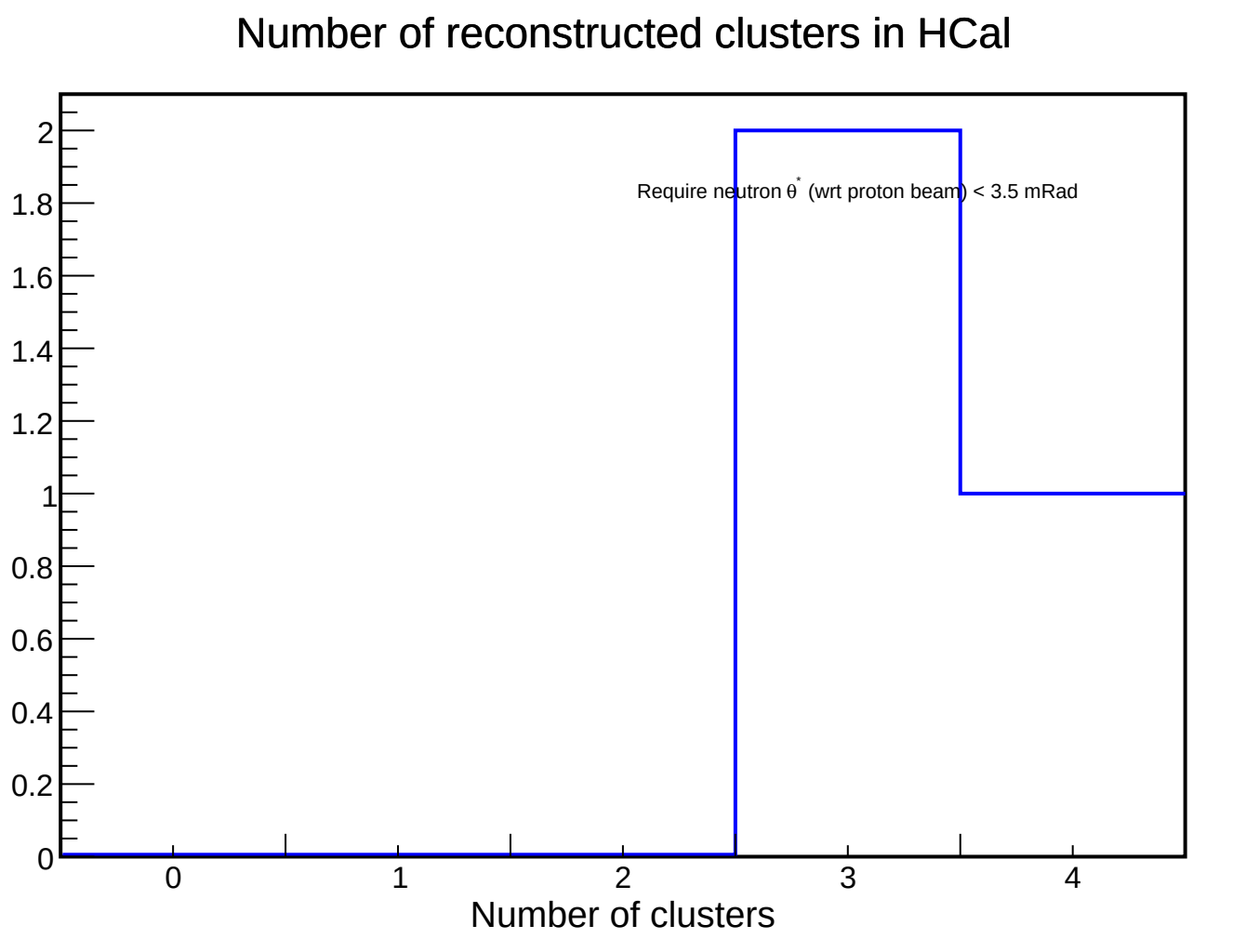
# Total reconstructed hit energy sum

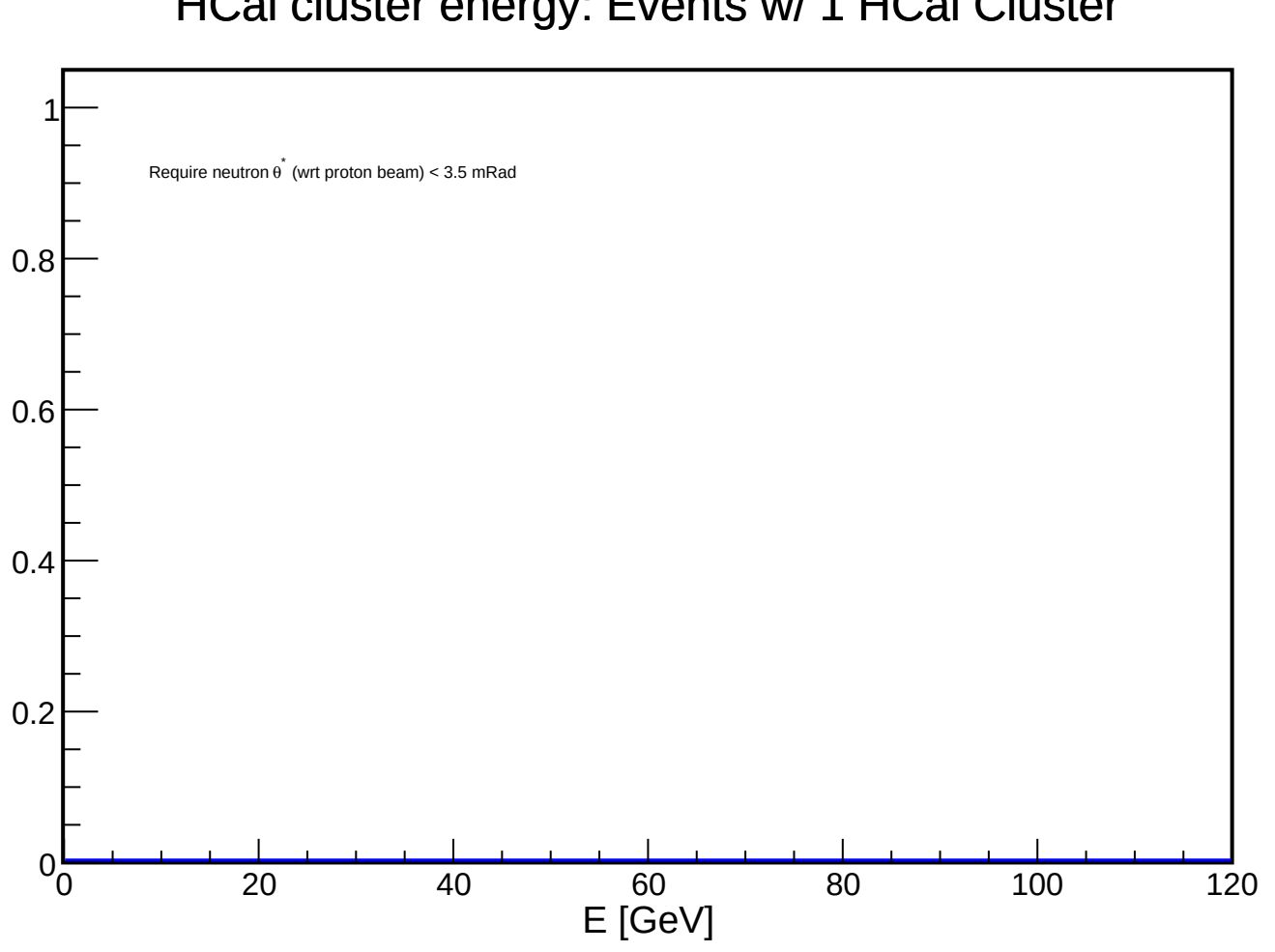


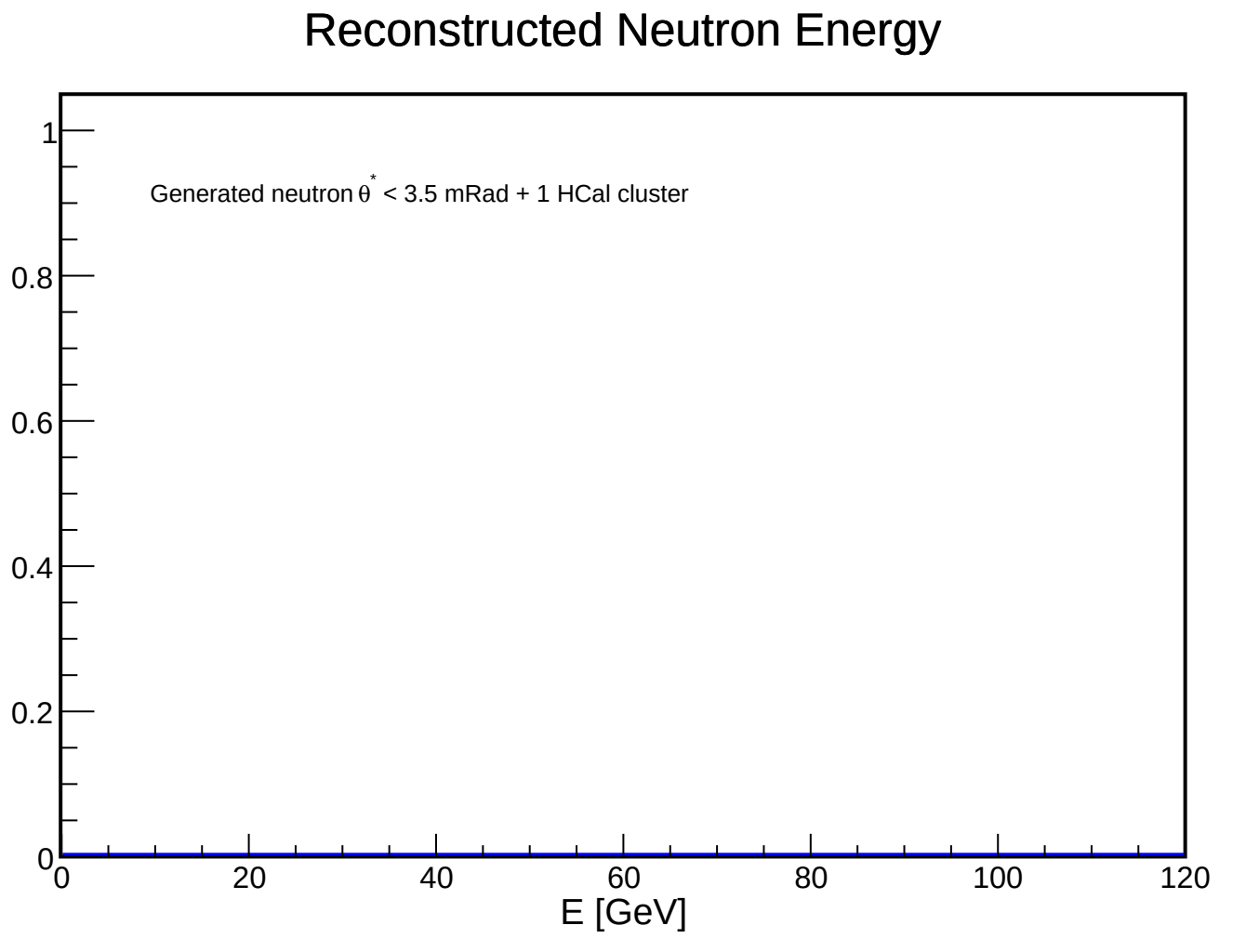
# Total reconstructed hit energy sum

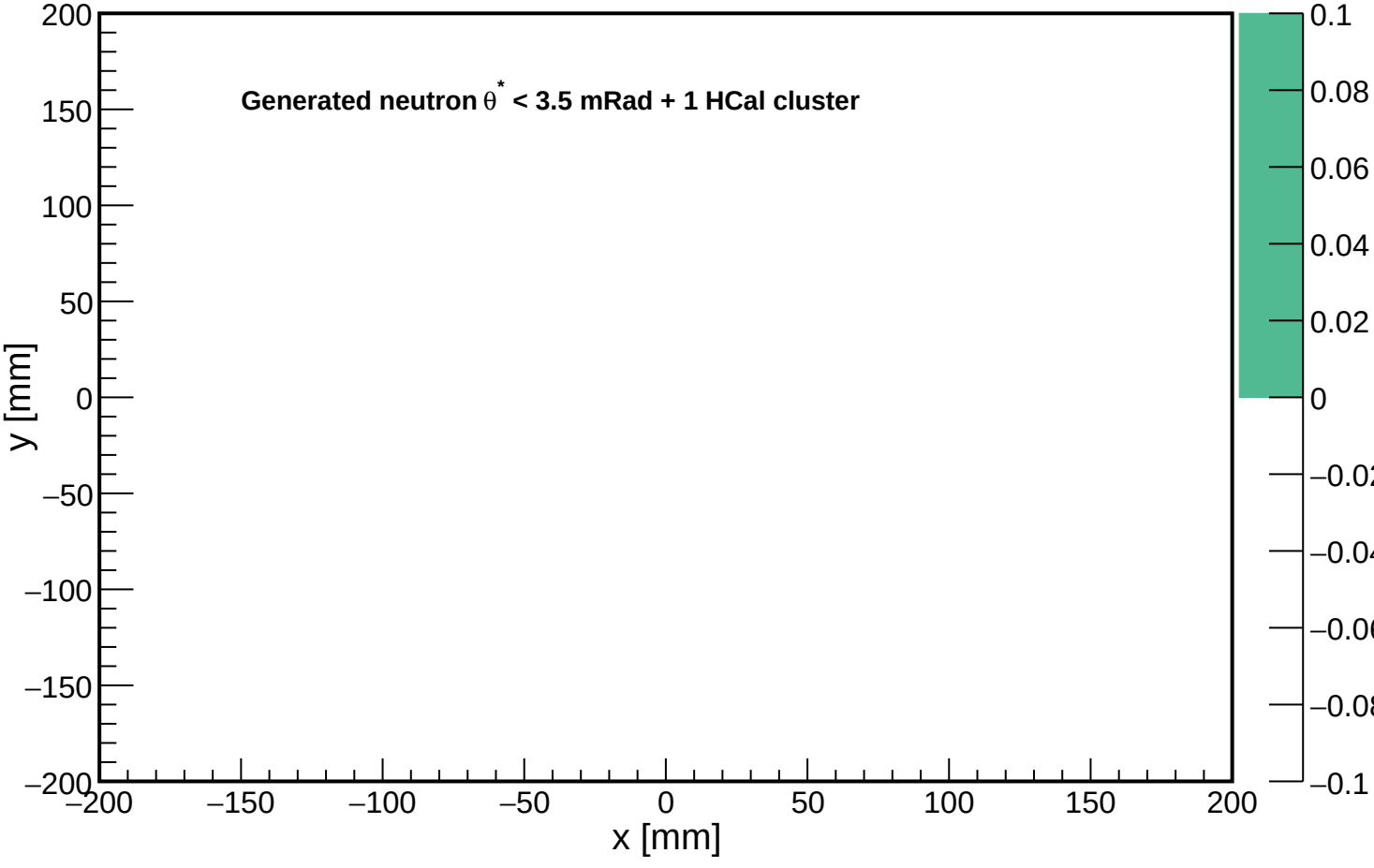



Number of reconstructed clusters in ECal
Require neutron θ* (wrt proton beam) < 3.5 mRad
600
500
400
300
200
100
0
0
1
2
3
4
Number of clusters


Number of reconstructed clusters in HCal
Require neutron θ* (wrt proton beam) < 3.5 mRad
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0
1
2
3
4
Number of clusters


HCal cluster energy: Events w/ 1 HCal Cluster
Require neutron θ* (wrt proton beam) < 3.5 mRad
1
0.8
0.6
0.4
0.2
0
0
20
40
60
80
100
120
E [GeV]


Reconstructed Neutron Energy
Generated neutron θ* < 3.5 mRad + 1 HCal cluster
1
0.8
0.6
0.4
0.2
0
0
20
40
60
80
100
120
E [GeV]


Reconstructed Neutron local hit position at ZDC HCal front face
Generated neutron θ* < 3.5 mRad + 1 HCal cluster
y [mm]
x [mm]
200
150
100
50
0
−50
−100
−150
−200
0.1
0.08
0.06
0.04
0.02
0
−0.02
−0.04
−0.06
−0.08
−0.1